MAKERERE UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS TECHNOLOGY

Data Communications BSE 2206

TIME: 10:00-11:00am 11th March, 2018

INSTRUCTIONS: ***Attempt all questions***

Time Allowed: One Hour

Question 1

a) Define Latency and state the four components of latency **(4marks)**

Latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source and the four components; propagation time, transmission time, queuing time, processing delay

b) Thermal noise is one of the types of noise and it can also be referred to as white noise. Explain why it is called white noise and mention any three other types of noise. **(4marks)**

It's called white noise due to random movement of atoms

Types of Noise

a. Intermodulation
b. Impulse
c. Crosstalk
d. Next End crosstalk

c) We have a channel with 6 MHz bandwidth. If we want to send data at 0.012 Gbps, Calculate the number of signal levels, SNR and SNR_{dB} **(6marks)**

$C = W\log_2 (1 + S/N)$ bits/sec

$12{,}000{,}000=6{,}000{,}000\log_2 (1 + S/N)$

$2= \log_2 (1 + S/N)$

$S/N=2^2-1$

S/N=3

$SNR = 10\log(S/N)$ dB

$SNR=10\log(3)$

SNR=4.8dB

$C=2W\log_2 L$

$12{,}000{,}000=6{,}000{,}000x2\log_2 L$

$12/12=\log_2 L$

$L=2^1$

L=2Levels

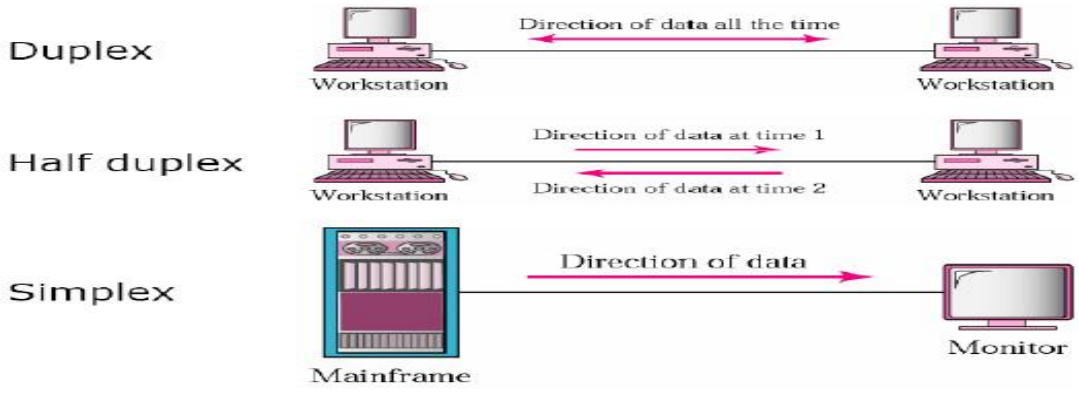
d) A non-periodic composite signal has a bandwidth of 40 MHz, with a middle frequency of 60MHz and peak amplitude of 10 V. The two extreme frequencies have an amplitude of zero and 5V respectively. Draw the frequency domain of the signal. **(6marks)**

Question 2

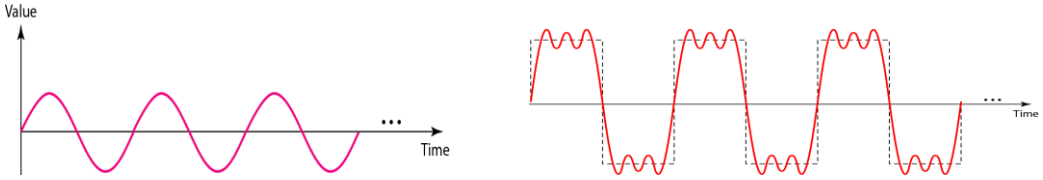
a. What is data communication and mention the four fundamental characteristics of effective data communication? **(3marks)**

Data communications: Exchange of data btn two devices via some form of transmission medium such as a wire cable. And the four characteristics are Delivery, accuracy, timeliness, and jitter.

b. Briefly explain the three modes of data transmission **(3 marks)**



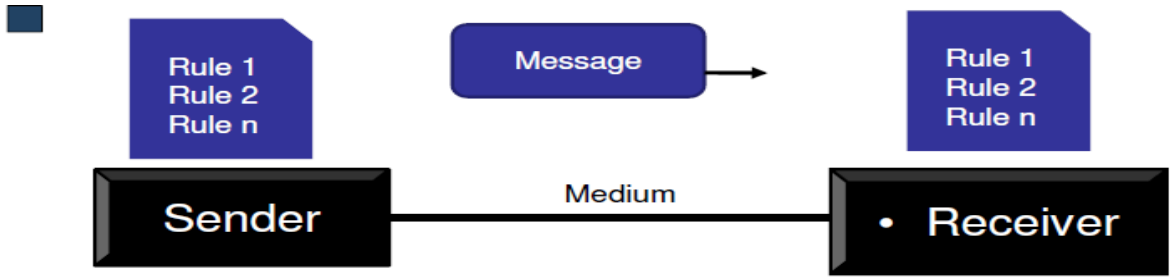
c. With aid of a diagram explain the difference between simple and composite signal **(2marks)**



Simple signal **Composite signal**

A simple analog signal cannot be decomposed into simpler signals. A composite analog signal is composed of multiple sine waves.

d. With aid of a diagram explain the five components of a communication system**(5marks)**



Message. Is the information (data) to be communicated. (text, numbers, pictures, audio, and video).
2. **Sender.** The device that sends the data message. It can be a computer. (workstation, telephoı handset, video camera, and so on).
3. ***Receiver.*** The device that receives the message. It can be a computer, workstation, telephone handset, television, etc.
4. **Transmission medium.** Is the physical path by which a message travels from sender to receiver. Twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol.** Is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

e. The period of a signal is 100µs and assume the speed of light is 3.0×10^8m/s. Calculate the wavelength of the signal in cm **(3marks)**

T=1/F T=100×10^{-6}=1/10000

F=1/1/10000=10,000Hz

$\lambda f = c$

$\lambda = 3.0 \times 10^8/10000$

$\lambda = 3.0 \times 10^4$m

In cm it will be 3.0×10^6 since 1m=100cm

f. The attenuation of a signal is -20 dB. What is the final signal power if it was originally 5W? **(4marks)**

$$\text{Attenuation (dB)} = 10 \times \log_{10}\left(\frac{\text{Input intensity (W)}}{\text{Output intensity (W)}}\right)$$

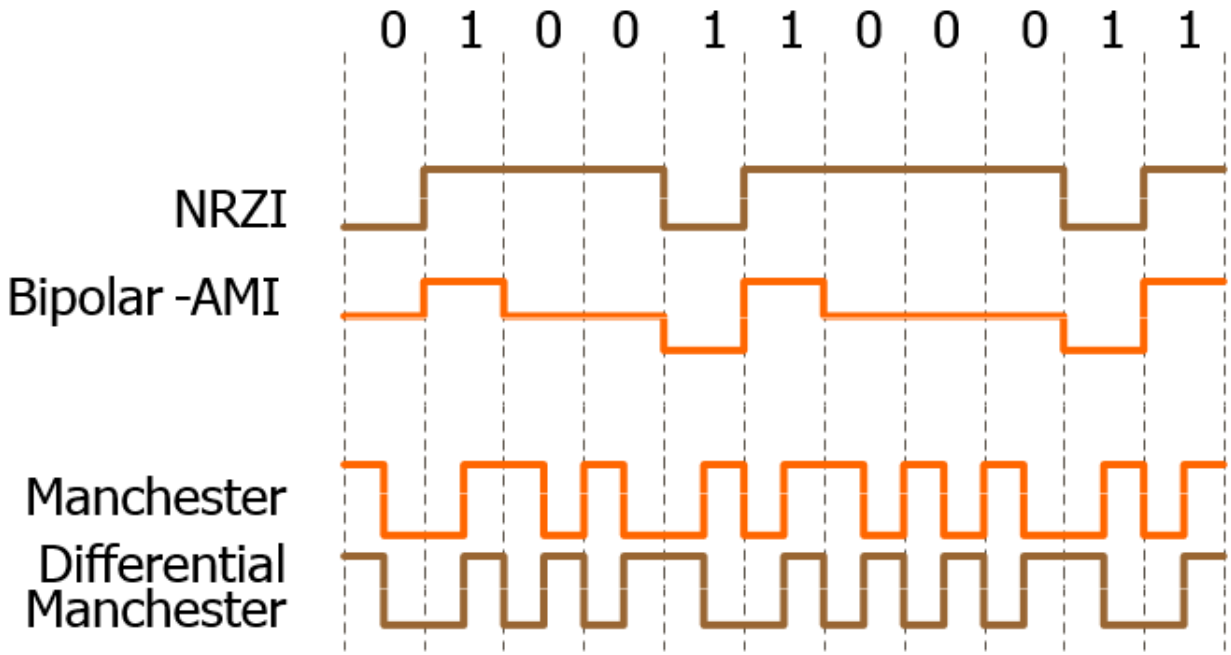
Let Y be the final power

-20 = 10log(Y/5)

-20/10 = log(Y/5)

-2= log(Y/5)

10^{-2} =Y/5

Y=0.05Watts

g. The signal in the diagram above is a NRZ-I. Decode the signal and encode the output bits using, Bipolar –AMI, Manchester and Differential Manchester **(10marks)**